

# Worksheet 1: Quadratic Equations (Mixed - 30 Problems)

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Class: Matric

Date: \_\_\_\_\_ Name: \_\_\_\_\_

**Instructions:** Solve the following quadratic problems. Show all steps clearly.

1. Solve:  $x^2 + 5x + 6 = 0$
2. Solve:  $2x^2 - 7x + 3 = 0$
3. Find the discriminant of:  $x^2 - 4x + 1 = 0$
4. Determine the nature of roots for:  $3x^2 + 2x + 5 = 0$
5. Solve:  $x^2 - 9x + 20 = 0$
6. Find the roots using the quadratic formula:  $2x^2 + 3x - 2 = 0$
7. Solve:  $3x^2 - 5x - 2 = 0$
8. Find the discriminant and nature of roots:  $x^2 + 2x + 3 = 0$
9. Solve for x:  $x^2 = 5x - 6$
10. Find the value of k if one root of  $x^2 + kx + 9 = 0$  is 3
11. Solve:  $4x^2 - 12x + 9 = 0$
12. Find the sum and product of roots for:  $x^2 - 6x + 5 = 0$
13. Word Problem: The product of two consecutive integers is 56. Form a quadratic equation and solve it.
14. Solve:  $5x^2 - x - 1 = 0$
15. Find the discriminant of:  $2x^2 - 4x + 2 = 0$
16. Determine if roots are real or imaginary:  $x^2 + 6x + 10 = 0$
17. Solve:  $6x^2 + 5x - 4 = 0$
18. Solve using factorization:  $x^2 - x - 12 = 0$
19. Word Problem: The area of a rectangle is  $48 \text{ m}^2$ . If its length is  $(x + 4) \text{ m}$  and width is  $(x - 2) \text{ m}$ , form and solve a quadratic equation.
20. Solve:  $x^2 - 10x + 25 = 0$
21. Find the discriminant and type of roots for:  $5x^2 + 4x + 1 = 0$
22. Solve:  $x^2 + 7x + 12 = 0$
23. Word Problem: A ball is thrown upwards with velocity  $14 \text{ m/s}$  from a height of  $2 \text{ m}$ . Its height  $h$  after  $t$  seconds is  $h = -5t^2 + 14t + 2$ . Find when it hits the ground.

24. Solve for  $x$ :  $9x^2 - 25 = 0$

25. Determine the nature of roots:  $2x^2 + 3x + 5 = 0$

26. Solve using the quadratic formula:  $3x^2 + 2x - 8 = 0$

27. Find the sum and product of roots for:  $x^2 - 3x - 10 = 0$

28. Solve:  $x^2 - 4x - 12 = 0$

29. Find  $k$  if one root of  $x^2 - 2kx + (k^2 - 5) = 0$  is  $(k + 1)$

30. Word Problem: The sum of a number and its reciprocal is 5. Form and solve a quadratic equation.